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Saveetha Institute of Medical And Technical Sciences
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TECHNICAL PRESENTATION

CSA0401:OPERATING SYSTEMS

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Real time traffic signal control architecture and OS design

1. OS Selection and Real Time Requirements



Choosing a preemptive RTOS for deterministic signal timing

Hardware abstraction and timing assurance

- *Use a board-support package with hardware timers and GPIO drivers so the OS can provide stable time bases for phase control.*



Justifying scheduling policy to meet green phase deadlines

Priority mapping from timing criticality

Assign higher priorities to signal actuation and timing supervision, while lower-priority tasks handle diagnostics and communication.

Overrun monitoring and deterministic recovery

Include deadline tracking, overrun counters, and controlled fail-safe behavior when tasks exceed their budget.



Scheduling policy aligned with phase windows

Apply fixed-priority preemptive scheduling or EDF to ensure each intersection phase task meets its configured green window.

Handling periodic and event-driven workloads

Model periodic tick-based updates and event-triggered controller changes, ensuring worst-case execution fits within phase margins.



Balancing interrupt latency and task responsiveness in hardware

Timing system calls for phase control

Use `delay_until`, `vTaskDelayUntil`, `timer_set`, and `semaphore_timed_wait` to start phase transitions at exact time offsets.

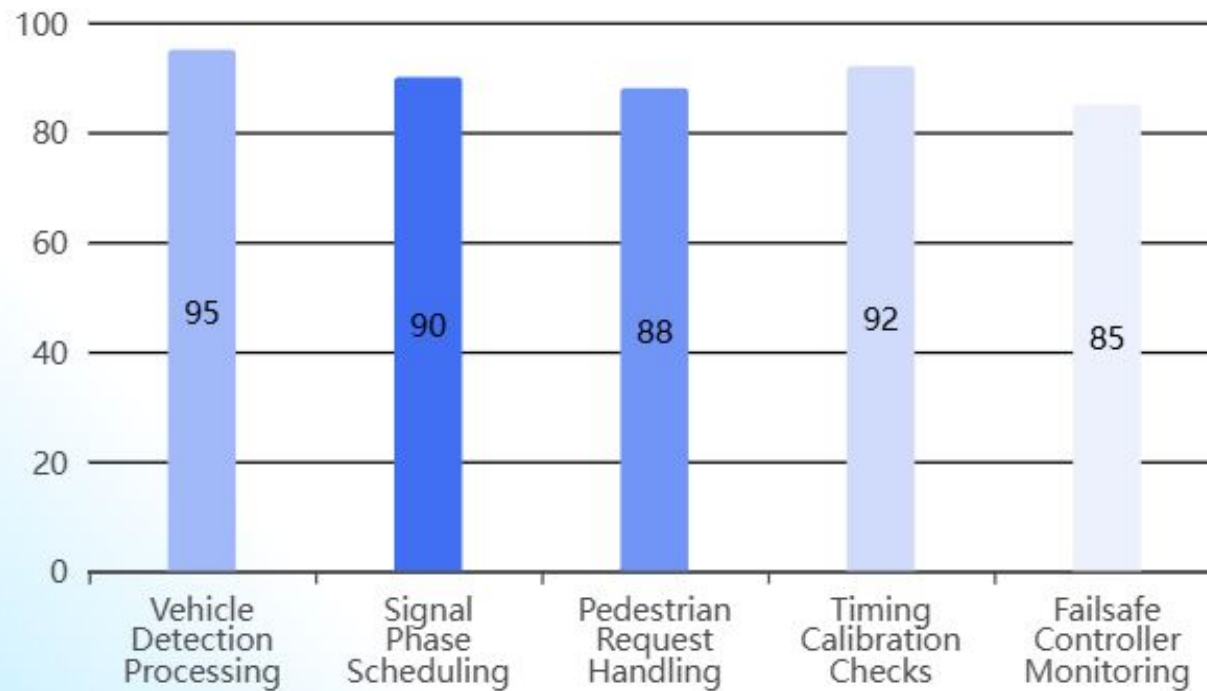


Interrupt-to-task latency budget management

Set interrupt priorities so timer capture and comms interrupts preempt background work without blocking critical OS scheduling events.



Real Time Traffic Signal Control Software



RTOS task set and timing services for traffic signal control(CPU Utilization)



2 . System Structure, Components, and Timing Calls



Task breakdown for controller logic, sensor input, and actuation



Real-time OS choice for safety

Use a preemptive RTOS to guarantee bounded phase switching and sensor response under strict latency requirements.

Justification for deterministic scheduling

Deterministic priority scheduling supports worst-case timing analysis, enabling compliant operation across traffic scenarios.

Controller task responsibilities

Separate tasks handle phase plan execution, pedestrian requests, and coordination between conflicting movements.



Synchronization primitives for safe shared state across tasks

1

Event-driven signaling between tasks

Use event flags or semaphores to wake controller logic immediately when detector thresholds or timers expire.

2

Avoiding priority inversion risks

Enable priority inheritance/ceiling protocols so high-priority timing tasks are not blocked by lower-priority holders.

3

Data consistency and buffering

Employ double-buffered sensor snapshots or lock-free queues to provide consistent inputs without long critical sections.



Process control and timing system calls for phase control

Timing primitives for fixed phases

Use delayed waits, periodic tick scheduling, and absolute-time timers to start and end phases precisely.



Task creation and lifecycle control

Create periodic and event tasks with RTOS task spawn calls, then delete or suspend them during configuration changes.

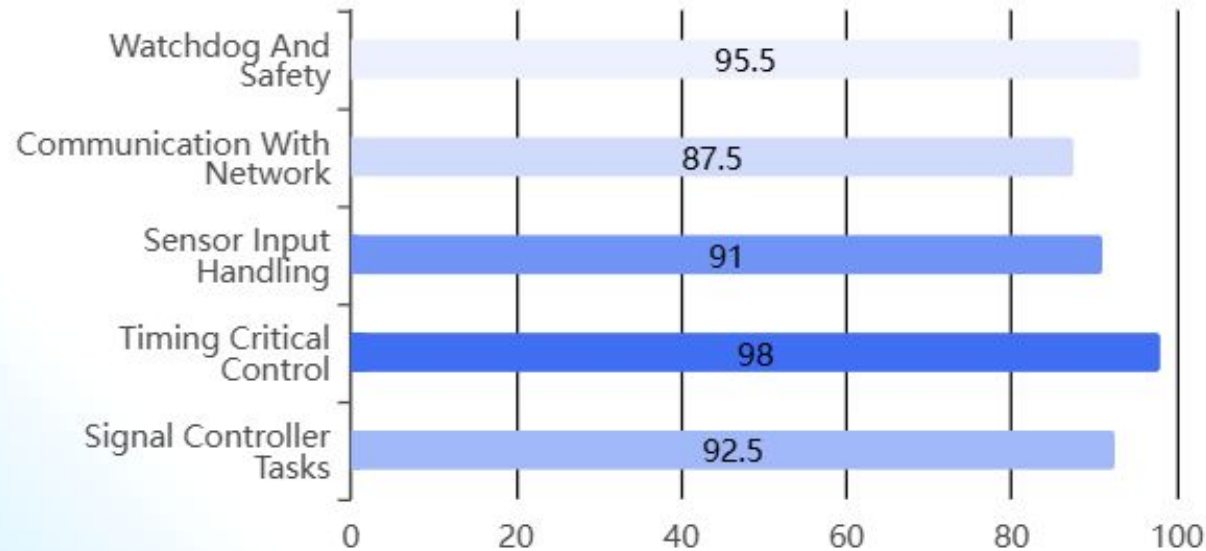


System calls for alarm and watchdog

Use timer callbacks and watchdog reset services to detect missed deadlines and force safe signal fallback.



Traffic Signal Embedded RTOS Overview



Real time traffic control OS for process control and timing(Percent



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Thank You